

BNI Insurance & Investment Services — Website

A modern, dark-themed, fully responsive website for **BNI Insurance & Investment Services**, built with a premium UI, smooth animations, a Node.js/Express backend, JSON file-based dynamic content management, and a full Admin Panel with CRUD functionality.

1. Project Overview

This project is a complete redesign and rebuild of the BNI Services website (reference: bniservices.in), delivered in two phases:

- **Phase 1 — Frontend:** A modern, dark-theme, animated, fully responsive multi-page website (Home, About, Services, Team, Testimonials, Contact) with premium UI components — gradient cards, hover effects, scroll-reveal animations, animated counters, a testimonial marquee, FAQ accordion, pricing cards, and a fully functional contact form.
- **Phase 2 — Backend:** A Node.js/Express REST API powering dynamic content (Services, Testimonials, Team) and contact-form submissions, plus a secure, JWT-authenticated **Admin Panel** to perform full CRUD (Create, Read, Update, Delete) operations on website content and view/manage incoming enquiries — all without touching code.

The goal was to exceed the visual quality and user experience of the reference website while keeping the codebase clean, lightweight, and easy to extend with a real database (MySQL/MongoDB) in the future.

2. Technology Used

Layer	Technology
Frontend Markup/Styling	HTML5, CSS3 (custom, no framework), Google Fonts (Poppins)
Frontend Scripting	Vanilla JavaScript (ES6+), IntersectionObserver API for scroll animations
Backend Runtime	Node.js
Backend Framework	Express.js
Authentication	JSON Web Tokens (JWT) via jsonwebtoken
Data Storage	JSON file-based data store (/backend/data/*.json) acting as a lightweight database
Middleware	cors, body-parser, express.json()
Environment Config	dotenv
Admin Panel UI	HTML/CSS/JavaScript (dark theme matching main site)

Note: The project currently uses a JSON-file data layer for simplicity and zero-setup evaluation. It is structured so it can be swapped for MongoDB/MySQL with minimal changes (see

| “Database Setup” below).

3. Installation Steps

Prerequisites

- Node.js v16 or higher installed
- npm (comes bundled with Node.js)
- A code editor (VS Code recommended)

Steps

1. **Extract the ZIP file** to your desired location.
2. Open a terminal and navigate to the backend folder:

```
cd bni-services/backend
```

3. Install the backend dependencies:

```
npm install
```

4. Create your environment file by copying the example:

```
cp .env.example .env
```

(On Windows: copy .env.example .env)

5. Open .env and adjust values if needed (default values work out of the box for local testing).
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4. How to Run the Project

1. From the backend folder, start the server:

```
npm start
```

For development with auto-restart on file changes:

```
npm run dev
```

2. Once running, you will see a confirmation in the terminal showing the active URLs.
3. Open your browser and visit:
 - **Website:** http://localhost:5000
 - **Admin Panel:** http://localhost:5000/admin
 - **API Base:** http://localhost:5000/api

The Express server serves the frontend, the admin panel, and the API all from a single port — no separate frontend server is required.

5. Database Setup

This project ships with a **JSON file-based database** located at backend/data/:

- services.json — list of services shown on the Home and Services pages

- testimonials.json — client testimonials
- team.json — team member records
- contacts.json — contact form submissions (populated at runtime)

No installation is required — these files are created and read automatically by the server.

Upgrading to a Real Database (Optional, for Production)

If you want to use MongoDB or MySQL instead of JSON files: 1. Install the relevant driver (mongoose for MongoDB or mysql2/sequelize for MySQL). 2. Replace the logic inside backend/utils/db.js (readData, writeData, nextId) with equivalent database queries. 3. Because all routes (backend/routes/*.js) only call these three helper functions, the rest of the application requires **no changes**.

6. Configuration / Important Files

File/Folder	Purpose
backend/server.js	Main entry point — starts the Express server and mounts all routes
backend/.env	Environment variables: port, JWT secret, admin credentials
backend/routes/auth.js	Admin login route, issues JWT tokens
backend/routes/services.js	CRUD API for Services
backend/routes/testimonials.js	CRUD API for Testimonials
backend/routes/team.js	CRUD API for Team Members
backend/routes/contact.js	Public contact form submission + admin view/manage
backend/middleware/auth.js	JWT verification middleware protecting admin-only routes
backend/utils/db.js	JSON file read/write helper (acts as the “database layer”)
backend/data/*.json	Data storage files
backend/public-admin/	Admin Panel UI (login + dashboard)
frontend/	All public website pages, CSS and JS

7. Features / Modules

Frontend

- Fully responsive design (Desktop, Tablet, Mobile)
- Dark, premium theme with purple/gold gradient accents
- Animated hero section with floating cards and live counters
- Scroll-reveal animations on all sections (IntersectionObserver)
- Auto-scrolling testimonial marquee
- FAQ accordion
- Pricing plan cards
- Sticky, glassmorphism navbar with mobile hamburger menu
- Functional, validated contact form connected to the backend API
- Pages: Home, About, Services, Team, Testimonials, Contact

Backend / Admin Panel

- RESTful API with full CRUD for Services, Testimonials, Team Members
 - Secure JWT-based admin authentication
 - Admin Dashboard with:
 - Overview statistics (total services, testimonials, team members, enquiries)
 - Add / Edit / Delete Services
 - Add / Edit / Delete Testimonials
 - Add / Edit / Delete Team Members
 - View, mark-resolved, and delete Contact Form submissions
 - Public contact form submissions stored and retrievable by admin
 - Dynamic content loading: Home and Services pages fetch live data from the API
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8. Things to Update After Setup

Before going live, please update the following:

1. **Admin credentials** — change `ADMIN_EMAIL` and `ADMIN_PASSWORD` in `backend/.env`.
 2. **JWT secret** — change `JWT_SECRET` in `backend/.env` to a long, random string.
 3. **Contact details** — update phone number, email, and address in the footer of each HTML page and in `contact.html`.
 4. **Company social media links** — update the placeholder links (#) in the footer's social icons.
 5. **Google Map embed** — replace the OpenStreetMap embed in `contact.html` with your exact business location.
 6. **Logo / favicon** — replace `frontend/images/logo.png` if you have an updated logo file.
 7. **Database** — for production, migrate from JSON files to MongoDB/MySQL as described in Section 5.
 8. **Domain & HTTPS** — configure a custom domain and SSL certificate when deploying to a live server (e.g., Render, Railway, VPS, or shared hosting with Node support).
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9. Admin Login Details

Field	Value
Admin Panel URL	<code>http://localhost:5000/admin</code>
Email	<code>admin@bniservices.in</code>
Password	<code>Admin@123</code>

⚠ These are **default development credentials** defined in `backend/.env.example`. Please change them immediately in your `.env` file before any real/production deployment.

10. Developer / Submission Details

Field	Detail
Full Name	Kshitij Jha
Contact Number	+91 7004655092
College Name	University of Petroleum and Energy Studies (UPES)

11. Folder Structure

```
bni-services/
├── backend/
│   ├── data/
│   │   ├── services.json
│   │   ├── testimonials.json
│   │   ├── team.json
│   │   └── contacts.json
│   ├── middleware/
│   │   └── auth.js
│   ├── public-admin/
│   │   ├── index.html      (Admin Login)
│   │   ├── dashboard.html  (Admin Dashboard)
│   │   └── admin.css
│   ├── routes/
│   │   ├── auth.js
│   │   ├── services.js
│   │   ├── testimonials.js
│   │   ├── team.js
│   │   └── contact.js
│   ├── utils/
│   │   └── db.js
│   ├── .env.example
│   ├── package.json
│   └── server.js
├── frontend/
│   ├── css/style.css
│   ├── js/script.js
│   ├── images/logo.png
│   ├── index.html
│   ├── about.html
│   ├── services.html
│   ├── team.html
│   ├── testimonials.html
│   └── contact.html
└── README.md
```

12. Notes

- All animations are implemented using pure CSS and the IntersectionObserver API — no third-party animation library required.
 - The site degrades gracefully: if the backend is not running, static fallback content from the HTML still displays correctly; only dynamic admin-managed content and the contact form require the backend.
 - Tested locally on Node.js v18+ with Chrome, Firefox, and Edge across desktop and mobile viewport sizes.
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Thank you for reviewing this project.